

Real options analysis of seawater desalination with solar photovoltaic energy systems: A least-squares stochastic perspective

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HIGHLIGHTS

- The authors employ a Least Square Monte-Carlo Real Options Analysis (LSMC-ROA) framework to evaluate the defer in investment of Photovoltaic-powered Reverse Osmosis (PV-RO) desalination plants under three different scenarios of powering RO.
- The study addresses a notable scarcity of similar studies that underscore the merits of applying ROA in freshwater production using renewable energy.
- To account for the uncertainties in water pricing, the study investigates the influence of price volatility and drift rates.
- The methodology was applied to a real-world case study of an RO desalination plant coupled with a photovoltaic power plant in the Kingdom of Saudi Arabia.

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ABSTRACT

Real Options Analysis (ROA) offers a modern theoretical framework to evaluate the impact of managerial flexibilities and uncertainties on the intrinsic investment value of renewable-powered desalination projects. This study analyzes Photovoltaic-powered Reverse Osmosis (PV-RO) desalination plants using a novel Least Square Monte-Carlo Real Options Analysis (LSMC-ROA) to assess the deferral of investment decisions. Three scenarios for powering RO are explored: reliance on the national grid with a subsidized electricity tariff for of \$0.085/kWh, which applicable only to governmental sections in Saudi Arabia, full PV power with grid balancing (\$0.083/kWh), and PV with a battery storage system (average electricity cost of \$0.138/kWh). The presented modeling results reveal that considering the flexibility to defer and wait for new information, the total investment values would be \$43.9 million, \$59.6 million, and \$30.5 million, respectively. Therefore, the optimal investment timing might be to defer investment to the second year. Additionally, the analysis reveals an unexpected outcome: a threshold salinity level for feedwater exists that can affect the economic viability of renewable-powered desalination. The proposed LSMC-ROA framework emerges as a robust tool for addressing uncertainties and assessing economic feasibility in the context of PV-RO desalination projects.

Nomenclature

Acronyms		Greek symbols	
CAPEX	capital cost of expenditure	Π	Osmotic pressure (bar)
DCF	Discounted Cash Flows	ΔP	pre-treatment pressure drop (bar)
DEEP	Desalination Economic Evaluation Program	α	Drift rate
DTA	Decision tree analysis	ϵ	Random value
		E	Expectation value

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GBM	Geometric Brownian Motion	η	efficiency of PV modules
KSA	Kingdom of Saudi Arabia	η	Efficiency (%)
LCOE	Levelized Cost of Electricity	σ	Volatility rate
LCOW	Levelized Cost of Water	ω	Simulation path
LSMC-ROA	Least Square Monte-Carlo Real Options Analysis	Subscripts, superscripts	
MED	Multiple Effect Distillation	Aux	Auxiliary power consumed
MSF	Multiple Stage Flash	b	Rejected brine

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NPV	Net present value	cmm	membrane replacement cost
OPEX	operation and maintenance costs	Cons	Construction period (yr)
PV	Photovoltaic	d	Desalinated
PV-RO	Photovoltaic-powered Reverse Osmosis	f	Feedwater
RO	Reverse osmosis	HP	High-head pump pressure rise
ROA	Real options analysis	Life	Seawater desalination plant life (yr)
ROV	Flexible options value	NDP	Net drive pressure
SMR	Small modular reactor	Pms	Annual water produced
WACC	weighted average capital cost	RET	Energy recovery
Roman symbols		SP	Seawater Pump
A	Surface area (m ²)	v	Validity time (yr)
AFWP	Availability factor of water plant (%)		
cmm	Specific membrane replacement cost (\$/m ³)		
Cons	Specific construction costs (\$/m ³ /day)		
dW	Wiener process		
E	Specific power of pumps (kW/m ³)		
f	performance ratio of PV modules		
GHI	annual in-plane global irradiance		
Inv	Investment cost (\$/m ³ /day)		
Inv	Investment cost (\$/m ³ /day)		
J	Ratio of nominal permeate to design permeate flux		
kmc	Water plant cost contingency factor (%)		
Kmo	Water plant owners cost factor (%)		
LTWP	Lifetime of water plant (yr)		
M	Membrane flow (m ³ /day)		
O&M	Operation and Maintenance cost (\$/m ³ /day)		
ON	Overnight capital cost (M\$)		
OV	Optio-to-deffer valus (M\$)		
P	Maximum design pressure of membrane (bar)		
q	electricity energy provided by non-renewable source (KW)		
r	Discount rate (%)		
Rr	recovery ratio		
SCF	Salinity correction factor		
T	Temperature (°C)		
t, θ	Time (yr)		
Tax	non-renewable energy tax expenditure (M\$)		
TCF	Temperature correction factor		
U	Uncertain variable		
V	Investment value (M\$)		
W	Power of pumps (MW)		
Wdr	Desalination plant capacity (m ³ /day)		
WPA	Water Plant A		
WSP	Water selling price (\$/m ³)		
X	Salinity (ppm)		
Φ	Cash flow (\$/yr)		

1. Introduction

Arid and semi-arid regions, particularly in the Middle East, face escalating freshwater scarcity, highlighting the urgent need for sustainable solutions. Seawater desalination presents a promising technology to address this growing crisis. However, its energy-intensive nature necessitates innovative approaches that integrate efficiency, sustainability, and economic viability. Current advancements in desalination technology prioritize reduced energy consumption and cost while minimizing environmental impact.

The increasing demands for both global energy and freshwater resources necessitate a focus on promoting renewable energy utilization. This approach aims to lessen dependence on non-fossil fuel energy sources and mitigate environmental concerns. Presently, the total global desalination capacity is around 100 million m³/day with an estimated 16,000–20,000 operational plants distributed across over 120 countries

[1,2]. This signifies a substantial increase, with a nearly 50 % rise in global desalination capacity from 66.4 million m³/day to 100 million m³/day observed within the past six years. This indicates the importance of this industry in arid and semi-arid areas such as the Middle East and North Africa region, which accounts for approximately 60 % of global desalination capacity [1].

Out of 80.0 % of the world's desalination capacity utilizes membrane process, with reverse osmosis (RO) being the leading mature industrial technology [3]. While seawater desalination remains an energy-intensive and expensive process compared to conventional freshwater treatment, RO desalination costs are decreasing, while the costs for developing new freshwater sources of potable supply are increasing or no longer possible [4]. Data provided by Mahmoudi and Panagopoulos [2,5] highlight the growing status of commercial RO desalination techniques powered by renewable energy around the world. Among these, RO powered by a solar photovoltaic system (PV-RO) is considered one of the most promising and holds the largest share of desalination systems (Fig. 1). However, the current implementation of PV-RO using a direct electrical technique where the components function separately offers significant room for improvement [6–8].

For instance, PV-RO desalination is particularly suitable for remote villages with insufficient rainfall, where scarce fossil fuel resources limit the use of traditional power generation [9].

Despite advancements in developing smooth-functioning, ambient temperature operating, and capacity-scalable PV-RO desalination plants [10], investments in this technology haven't grown as anticipated. The lack of robust economic justification for using renewable energy resources remains a key hurdle. Another example, PV/T collectors offer advantages over PV modules by enhancing both electrical and thermal yield per unit area. [11,12] They also demonstrate higher productivity during summer and efficiency during winter [13].

Another challenge lies in the trade-off between RO desalination performance, measured by desalinated water quantity, and the limitations imposed by battery storage capacity. While Ghamrawi et al. [14] proposed a novel flow-tracking technique to adapt seawater intake to solar irradiance and power generation, eliminating the need for battery storage in PV-RO systems, further research is needed.

For decision-makers, the main parameters for investing in PV-RO desalination plants are total investment costs, produced water costs, and site-specific factors like local incentives and subsidies. The estimated cost of desalination is highly site-dependent and hinges on the chosen energy source, whether it be fossil fuels, nuclear [15], or

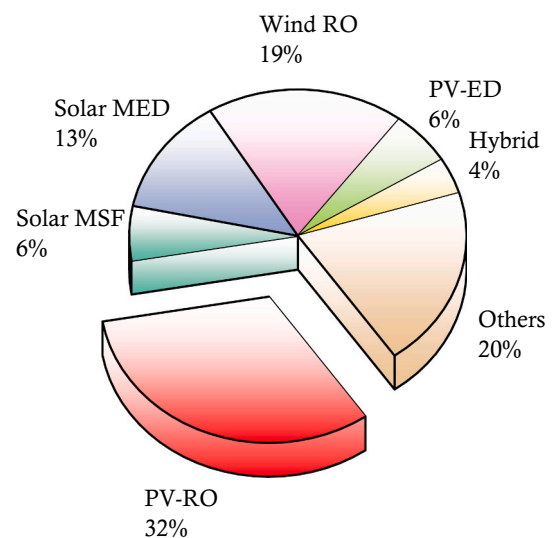


Fig. 1. Distribution of different commercial desalination techniques powered by renewable energy sources [1].

renewable energy to power desalination technology.

Traditional valuation tools, including, the Discounted Cash Flows (DCF) and decision tree analysis (DTA) [16] have been widely used for investment evaluation. However, these methods often struggle to account for the inherent uncertainties and managerial flexibilities associated with real-world projects, particularly those involving renewable energy technologies. This study addresses this gap by employing Real Options Analysis (ROA) to assess the economic attractiveness of investing in a renewable energy desalination system for seawater treatment [17–20], using the case of the Kingdom of Saudi Arabia (KSA). The basic motivation for this research is to utilize the advantages of Real Options Analysis (ROA) to calculate the embedded option values of (1) powering the RO plant using national grid, (2) employing the complete electricity output generated by a photovoltaic (PV) power plant, and relying on the national power grid to offset any surplus energy needs of the reverse osmosis (RO) plant, and (3) integrating a battery storage system to support the PV power plant during periods of higher demand than its production capacity. Subsequently, we assess the optimal timing of investment by computing the value associated with deferring the investment decision. Ultimately, we evaluate the impact of variations in the independent variables on investment determinations through a sensitivity analysis.

DCF and DTA methods are standard investment appraisal tools used by analysts and other professionals in project valuation, and they serve the purpose very well in many applications, DCF is a mathematical deterministic model, where cash flows are discounted to the current value, and the NPV or the underlying asset value of the project is obtained as the sum of DCF over the investment life cycle of the project. In industrial practice, DCF analysis is a deterministic model that is used as the most widely project appraisal method [21]. Although the DCF analysis is simple to implement, it has substantial drawbacks which are discussed in the following: The DCF analysis implicitly assumes that all future decisions are made today and couldn't capture the value of managerial flexibility during the project life cycle. While usually the investment decisions are continuously revised following the regulatory systems' decisions and market changes. There can be a range of possible outcomes over the life of a project, with uncertainty and volatility over time. Therefore, the future cash flows behave stochastically and cannot be assumed as certain. However, in DCF analysis uncertainties and risks analyses are based on forecasts or biased. As an illustration, when contemplating a conventional power plant with a life cycle of 40–50 years, where notable variations in electricity tariffs and fuel outlays can manifest. To tackle this predicament, prevalent methodologies encompass sensitivity analysis, scenario analysis, and Monte Carlo simulation. Regrettably, these techniques can, at times, be influenced by arbitrary assumptions dictated by circumstantial factors [22].

Despite DCF techniques, DTA is applicable when there is uncertainty about project outcomes and opportunity for contingent decisions exists, DTA could provide a strategic road map, depicting alternative decisions, their costs, their possible outcomes, and the probability and the payoff of the outcomes. DTA is a more sophisticated tool than DCF and offers value when a project is multistage and contingent decisions are involved.

Because of the limitations discussed above, the ROA offers a modern theoretical framework to evaluate appropriately the change in underlying project investment value under managerial flexibilities and uncertainties. Compared to the DCF and DTA analysis tools, the ROA method is mathematically more complex to implement. However, ROA is not a substitute for any of the DCF and DTA methods discussed. It uses DCF analysis as a building block and allows you to integrate DTA, as necessary, into a sophisticated framework that provides analysts and decision-makers with more meaningful information. ROA has been widely accepted by academics as a tool for capital budgeting and investment appraisal [21].

To date, certain challenges have hindered the extensive adoption of ROA within practical contexts. Despite ROA's pertinence within the

energy domain, its utilization in the realm of desalination remains relatively limited. For instance, Zhang and Babovic [20] adopted a decision support framework using a ROA approach by evaluating innovative water technologies from multiple perspectives under uncertainty to evaluate the design and architecture of Singapore's water supply systems to be more independent of the influence of the policies' government of Malaysia. Locatelli et al. [23] used the key advantages of the ROA method to demonstrate the complexity surrounding the decision about the investment of the load following the operation of a cogeneration Small modular reactor power plant. They investigate the option to defer and build and the option to switch between two distinct output products strongly depends on the relative pricing dynamics of water and electricity.

Nevertheless, energy utilities and stakeholders in the energy sector are very keen to work on the ROA framework as long as the results provide valuable indications supporting the decision-making process [21].

The main contributions of this study could be summarized as follows:

- Demonstrating the application of Real Options Analysis in solar photovoltaic energy-powered reverse osmosis desalination projects.
- Introducing a Least-Square Monte-Carlo of Real Options framework to emphasize the superiority of Real-Options Analysis over traditional DCF methods.
- Providing the decision makers with a prioritization framework for different design parameters.
- Applying the methodology to a real-world case study of an RO desalination plant coupled with a photovoltaic power plant in KSA.

The rest of this article is structured as follows. Section 2 describes the technical model of the RO desalination plant and the real options model as well as model solution procedures. In Section 3, we apply the proposed model to evaluate solar PV-powered RO desalination plants in KSA. The last section concludes this study with some policy implications.

2. Methodology

In this section, we outlined the detailed roadmap of the systematic approach to address the research objectives. In continue, the authors provide an in-depth exploration of techno-economic modeling of the RO desalination process, and second, the utilization of Real Option Analysis coupled with an advanced computational technique that empowers the authors to quantitatively evaluate real options within complex investment scenarios.

2.1. Reverse osmosis

Globally, two main types of desalination technologies are in use, which are membrane desalination processes, such as reverse osmosis, and thermal distillation, such as Multiple Effect Distillation (MED) and Multiple Stage Flash desalination (MSF). In the thermal process, feed-water is boiled, and the vapor is condensed as pure water (distillate), while in the RO process, seawater with a high concentration of salts is pumped by pressure through a semipermeable membrane of reverse osmosis to filter out dissolved solids and obtain water of low concentration of salts. An RO desalination process consists of three main components: osmotic pressure, salt rejection, and the recovery ratio. An RO desalination plant consists mainly of a high-pressure pump and a membrane module, as shown in the schematic representation of Fig. 2.

Almost all modern large-scale RO facilities use power recovery turbines, where the pressure of the discharged brine stream is utilized to reduce the system's overall energy consumption. The mathematical modeling of the RO unit is based on the details taken from the

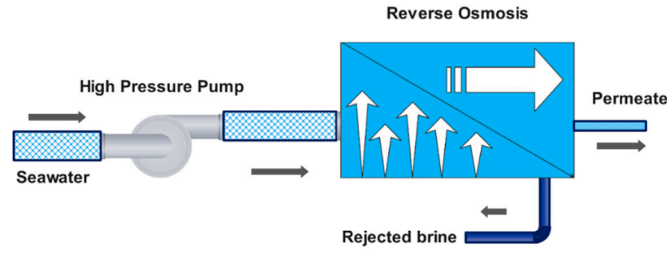


Fig. 2. Basic schematic of an RO unit [15].

Desalination Economic Evaluation Program (DEEP) software developed by the International Atomic Energy Agency [24,25]. The following assumptions are considered to model the RO unit in MATLAB software:

- The operation is conducted under constant temperature conditions.
- No chemical reactions occur throughout the desalination process.
- The density of seawater is held at a constant value of 1000 kg/m³.
- The absolute pressure at the freshwater outlet is presumed to be 1 bar.

The permeability coefficients of water and various salts within each membrane are held constant.

Eqs. (1)–(5) demonstrate the total power consumption of the RO unit, denoted as W_{RO} , which is comprised of various components, including the energy required by its high-pressure pumps (W_{HP}), the power consumed by the seawater pump (W_{SP}), the energy recovered during the process (W_{RET}), as well as additional auxiliary energy usage (W_{AUX}).

$$W_{RO} = W_{hp} + W_{sp} + W_{RET} + W_{aux} \quad (1)$$

$$W_{hp} = \frac{M_f \cdot \frac{1000}{24 \times 3600} \times P_{HP}}{98600 \times \eta_{hp}} \quad (2)$$

$$W_{sp} = \frac{M_f \cdot \frac{1000}{24 \times 3600} \times P_{sp}}{98600 \times \eta_{sp}} \quad (3)$$

$$W_{RET} = [-(1 - Rr) \times E_{RET} \times W_{hp}] \quad (4)$$

$$W_{aux} = W_{dr_{RO}} \times \frac{E_{aux}}{24 \times 1000} \quad (5)$$

The model developed for the RO unit takes inputs as shown Table 1. Table 2 details the formulation used in the model.

In the context of this RO facility, it operates continuously year-round. As documented in prior research [26], the Availability factor of the water plant (AFWP) parameter has been noted to extend up to 90 %, resulting in an annual total water production of $(10,000 \text{ m}^3/\text{day} \times 365 \text{ day} \times \text{AFWP})$ 3,285,000 m³/year. Corresponding to the availability of the RO plant, the yearly energy consumption amounts to $(1.4 \times 10^3 \text{ Kw} \times 24 \text{ h} \times 365 \text{ day})$ 12,226,400 kWh/year, with a specific energy

consumption rate of $(12,226,400 \text{ kWh} / 3,285,000 \text{ m}^3/\text{day})$ 3.36 kWh/m³/day.

3. Real-Options-Analysis (ROA) model

Consider a scenario in which decision-makers are confronted with the task of determining the optimal timing for the implementation of a renewable desalination venture. As depicted in Fig. 3 we present a visual representation of the potential sequence of cash flows, incorporating both incoming and outgoing funds over consecutive periods. These periods span various stages, encompassing both the developmental and operational phases characterizing a typical project associated with a power plant.

The yearly cash flow of an RO desalination plant comprises the returns from selling. Freshwater, operation and maintenance costs, and tax expenditures. We could add the gains from the tax by precluding CO₂ pollution into the atmosphere in the form of CO₂ emission allowances to the income cash flow. The investment in reverse osmosis desalination plants can be irreversibly initiated in the year t through the validity period $(1 \leq t \leq t_v)$. Then, the production, and sale of produced fresh water begins immediately after the project completion at the year $t + \theta_{cons}$, and continues for the unit's lifetime (θ_{Life}) until $t + \theta_{cons} + \theta_{Life}$.

In this framework, ROA provides decision-makers with an opportunity to assess managerial flexibility as well as the effects of different risks. This problem could be solved by evaluating the value of the conditional expected payoff to option-holders from continuation using dynamic programming. An efficient technique to value the conditional expectation function involved in dynamic programming is the introduction of the Least-Square Monte-Carlo (LS-MC) method. This is accomplished through a process of regression analysis, where the forthcoming actual cash flows associated with continuing the operation are regressed against a set of basic functions representing the values of pertinent state variables. The resultant fitted value derived from this regression analysis stands as an efficient and unbiased estimate of the conditional expectation function. It, in turn, enables us to precisely gauge the optimal stopping criteria for the option. Implementation of stochastic linear regression due to uncertainties constitutes the LS-MC procedure's main difficulty. Schwartz developed a simple, yet powerful algorithm to overcome this difficulty [28].

The primary goal of the LS-MC technique is to provide an approximate path to the optimal stopping rule that maximizes the value of exercising an option. The LS-MC simulation method offers a more realistic modeling approach for capturing the pivotal uncertainties that impact the project's valuation, distinguishing it from closed-form methods like the Black-Scholes technique. This attribute renders it particularly well-suited for application within the context of power generation and investments in renewable energy.

However, reviewing the literature in the field reflects no similar use of ROA tools in desalination. Furthermore, the LSMC technique not only demonstrates effectiveness in appraising financial and real options but also can be extended to valuing complex real investments with several embedded real options and multiple variables characterized by uncertainty. For example, in the present framework, the authors place focus on determining the optimum investing time by postponing or delaying the investment until uncertainties are resolved.

3.1. The Least-Square Monte-Carlo Real Options Valuation model

The investment environment of renewable energy is closer to a stochastic scenario that is characterized by uncertainty and managerial flexibility. Therefore, the assessment starts with generating simulation paths, ω , to replicate the stochastic dynamics of different technical and economic state variables. Since the value of the produced fresh water by utilizing renewable energy is dependent on many uncertain factors, especially the renewable electricity cost in its lifetime, it is reasonable to

Table 1
Reverse osmosis desalination plant model data.

Variable	Description	Value
$W_{dr_{RO}}$	Desalination plant capacity (m ³ /day)	10,000
X_f	Feedwater salinity (ppm)	30,000–45,000
X_d	Produced water quality (ppm)	100–200
T_f	Feedwater temperature (°C)	35–45
T_d	Produced water temperature (°C)	35–45
AFWP	Availability factor of water plant (%)	90
LTWP	Lifetime of water plant (yr)	20

Table 2
Mathematical modeling of the RO unit.

Equations
$Rr = 1 - \frac{0.0115}{P_{MAX}} \cdot X_f$ (1)
Mass balance
$M_f = \frac{Wdr_{RO}}{Rr}$ $M_b = \left(\frac{1}{Rr} - 1\right) \cdot Wdr_{RO}$ (2)
Salinity balance
$X_d = 0.00125 \cdot X_f \cdot J_w \cdot \left(1 + \frac{1}{1 - Rr}\right) \cdot (1 + 0.03 \cdot (T_f - 25))$ (3)
$X_b = \frac{X_f}{1 - Rr}$ (4)
Energy balance
$A = A_{ref} \times TCF$ (5)
$TCF = \exp\left(\frac{2200}{8.31} \times \left(\frac{1}{T_f + 273.15} - \frac{1}{298}\right)\right)$ (6)
$\Pi_f = 0.0000348 \times (T_f + 273.15) \times X_f / 14.7$ (7)
$\Pi_b = 0.0000348 \times (T_f + 273.15) \times X_b / 14.7$ (8) $SCF = 1.5 - 0.000015 \times \frac{\Pi_f + \Pi_b}{2} \times \left(1 + \frac{1}{Rr - 1}\right) \times X_f$ (9)
$P_{NDP} = J_w \times SCF \times NDPN \times \frac{TCF}{FF}$ (10)
$P_{HP} = P_{NDP} + \frac{\Pi_f + \Pi_b}{2} + \Delta P$ (11)

Where P_{sp} is the seawater pump head in bar, E_{aux} is the specific auxiliary energy consumption in kW/m³, E_{RET} is the energy recovery efficiency factor, and η_{hp} and η_{sp} are the efficiencies of high-head pumps and seawater suction pumps, respectively. The model for the RO desalination was validated against the DEEP software and applied to an RO unit with a daily production capacity of 10,000 m³/day. The identical parameters and inputs employed in this study were inputted into the model, and the outcomes were compared, as presented in Table 3. As depicted in this table, for all scenarios, the margin of error remains below 8 %.

Table 3
Validation of the RO unit with capacity of 1000 m³/day using DEEP software.

Output parameter	Unit	Present study	DEEP software	Error (%)
Rr	–	23.9	24.0	0.4
P_{HP}	Bar	59.1	59.2	0.1
W_{hp}	MW	3.74	3.74	5.8
W_{sp}	MW	0.10	0.10	–
W_{RET}	MW	0.16	0.16	–
W_{aux}	MW	–2.54	–2.71	6.2
W_{RO}	MW	1.40	1.52	7.2

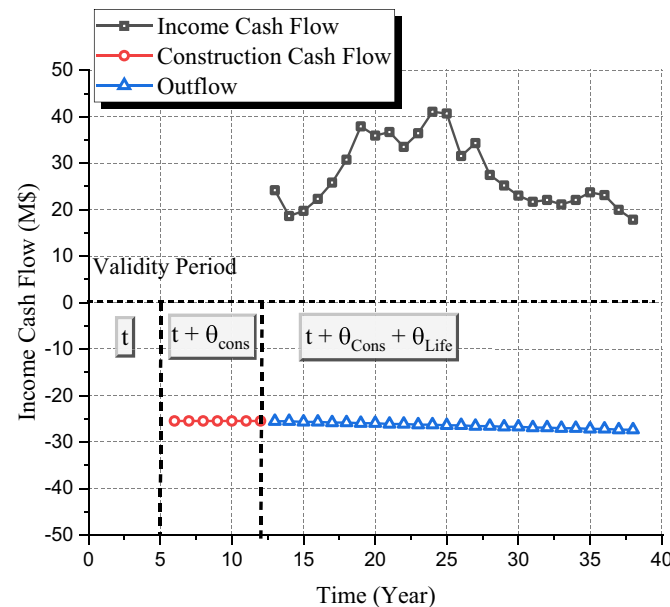


Fig. 3. Hypothetical chronology of levelized discrete cash flows of a power plant [27].

express the project value by its expectation $E[-]$.

Since, we could express the total net present value of the expected underlying asset during the validity period of the investment as follows [27]:

$$V(t) = E \left[\sum_{m=t+\theta_c}^{t+\theta_c+\theta_L} e^{-r \times (m-t-\theta_c-\theta_L)} \times \varphi_m - \sum_{n=t}^{t+\theta_c} e^{-r \times (n-t)} \times Inv_n \right] \quad (6)$$

where r is the discount rate, φ denotes the cash flow, and Inv , denotes the investment cost. Suppose that, WSP is the selling price of fresh water, Wdr is the capacity of the RO plant, q^{NRE} and P^c are the quantities of electricity energy provided by non-renewable sources for the reverse osmosis power plant and the price of CO₂ emissions, respectively. Further, $O\&M$, is the price of operating and maintenance costs and tax is the non-renewable energy tax expenditure. We then have:

$$\varphi_t = WSP \times Wdr_t^{RO} + q_t^{NRE} \times P^c - O\&M_t - tax \quad (7)$$

Suppose that the Levelized Cost of Water (LOCW) is the final water production cost which is calculated by dividing the annual required revenue of the RO desalination plant attributable to water production by the annual water production and comprised of the capital cost of expenditure (CAPEX), operation and maintenance costs (OPEX), and energy cost. WSP is considered the LCOW with the most expensive energy source. In this paper, the authors assume this value is adjusted based on the unsubsidized price for electricity energy.

3.1.1. Algorithm

The detailed proposed algorithm to evaluate the optimum investment timing of a PV–RO desalination plant is generally built on the contribution of Najafi and Talebi's work [27].

Technical and economic deterministic values of the RO water plant (i.e., size of the water plant, temperature and salinity of the feedwater, specific capital investment and O&M cost of the RO plant, discount rate, economic lifetime, capacity factor, interest rate, etc.) are defined based on historical data and assumptions. The key idea/advantage of the ROA method is to establish uncertainty factor for state variables. The stochastic process is an appropriate technique to introduce uncertainty into ROA analysis. The Geometric Brownian Motion (GBM) process is usually chosen to reflect longer-term uncertainty and is suitable for the energy

sector [29]. Therefore, in this study authors used GBM to describe the motion process for specific capital investment and O&M cost, and water selling price, RE cost, and selling water price are key factors influencing the pace of the renewable desalination development (or economic feasibility). Further, PERT distribution is used to describe the water plant availability.

In the first step, for any simulation path, ω , the developed techno-economic model of the RO desalination unit is used to evaluate the expected NPV of revenues and outcomes at each discrete decision point during the validity period (t_v).

In continuation, from the maturity date ($t = t_v$), and recursion progresses back-ward until the present day ($t = 1$) if the total present value of the revenue due to selling produced water is less than the “exercise threshold”, the option holder must defer the investment ($L_{ij} = 0$). On the contrary, if the total present value of the revenue is greater than the “exercise threshold” ($L_{ij} = 1$), the option holder could invest immediately. The value of the option-to-defer can be expressed as [27]:

$$F(t, \omega) = \max \left\{ V(t, \omega) - E \left[\sum_{m=t}^{\theta_c + \theta_o} e^{-r \times (m - \theta_c - \theta_o)} \times \varphi_m - \sum_{n=t}^{\theta_c} e^{-r \times n} \times \text{Inv}_n \right], 0 \right\} \quad (8)$$

An exercise threshold is a rule that gives investors the right to invest based on the values of one or more state variables. This entails the removal of government subsidies and the incorporation of international diesel prices [26] into our analysis. Consequently, electricity acquired from the grid would be marketed at the international rate of \$0.4 per kWh in Saudi Arabia.

At the time $t_v - 1$, if the expected value of the revenue due to selling produced water to the grid is less than the “exercise threshold” is better for the option holders to continue the option's life until the final expiration date ($L_{ij} = 0$, $\Gamma_{ij} = 0$). On the other hand, if the expected value of the revenue is greater than the “exercise threshold” the option holders could exercise the investment immediately ($L_{ij} = 1$, $\Gamma_{ij} = 1$). The optimum value of the option-to-defer can be exercised as:

$$F(t, \omega) = \max \left\{ V(t, \omega) - E \left[\sum_{m=t}^{\theta_c + \theta_o} e^{-r \times (m - \theta_c - \theta_o)} \times \varphi_m - \sum_{n=t}^{\theta_c} e^{-r \times n} \times \text{Inv}_n \right], e^{-r \times t} \times F(t+1, \omega) \right\} \quad (9)$$

where $e^{-r \times t} \times F(t+1, \omega)$ is the discounted back the conditional expected value from continuing which would be received at the next time step. The recursion continues by rolling back in time and repeating the algorithm until the exercise decisions at each decision point per path. The optimum expected value of the option-to-defer (OV) is calculated based on Eq. (10). Further, the optimum timing of investment (t_{op}) in path ω is calculated as follows [28]:

$$F = \begin{bmatrix} F_{1,1} & \cdots & F_{1,t_v-1} & F_{1,t_v} \\ \vdots & \vdots & \vdots & \vdots \\ F_{\omega,1} & & F_{\omega,t_v-1} & F_{\omega,t_v} \end{bmatrix}_{\omega \times t_v} \quad (10)$$

$$NPV(F) = \begin{bmatrix} NPV(F_{1,1}, \dots, F_{1,t_v}) \\ \vdots \\ NPV(F_{\omega,1}, \dots, F_{\omega,t_v}) \end{bmatrix}_{\omega \times 1}$$

$$OV|_{1 \times t} = \frac{1}{\omega} \sum NPV(F)$$

$$t_{op} = MOD(L|_{\omega \times t}) \quad 1 \leq t \leq t_v$$

$$t_{op} = MOD(L|_{\omega \times t}) \quad 1 \leq t \leq t_v \quad (11)$$

When the option-to-defer is exercised at time t , the cash flow in the next time steps becomes zero. This is because once the option is exercised, there are no future cash flows or the expected value of the continuation.

The economic viability of desalination using renewable energy is influenced by several uncertain factors, including water selling price (directly affected by electricity costs), RO plant-specific investment and O&M expenditures, RO plant capacity, water salinity, interest rate, CO₂ price, water plant availability, and so on. Under a market-based price mechanism, fluctuations in these factors determine the market price of produced water.

Several relevant studies have advocated the utilization of Geometric Brownian motion (GBM) to model the dynamic process. In this study, the stochastic process that follows the technological advancements and economic uncertainties must conform to the following differential equation:

$$dU_t = \alpha \cdot U_t \cdot dt + \sigma \cdot U_t \cdot dW_t \quad (12)$$

where U_t , denotes the uncertain variable at time t , α is drift rate, and σ is volatility rate. dW represents the Wiener process. The expected value U_t , could be formulated as follows:

$$E[U_t] = U_0 \cdot \exp \left(\left(\alpha - \frac{\sigma^2}{2} \right) \cdot dt + \sigma \cdot \sqrt{dt} \cdot \varepsilon \right) \quad (13)$$

where $\sigma \cdot \sqrt{dt}$ represents the Wiener process and ε is the normally distributed random variable with mean zero and unit variance.

3.1.2. Parameters estimation

This section provides the descriptions of the determination of parameters used in this study which are mainly collected or compiled through literature and presented in Table 4. Following the research methodology explained earlier the contribution of cash flow (φ) will be presented and explained in continuing.

3.1.2.1. Cost of produced water. The RO CAPEX or the total plant investment cost during the construction period (Inv) includes the

Table 4
Financial and technical parameters for stochastic variables.

Variable	Description	Initial value
Sp_{CONS}^{RO}	Specific construction cost of RO desalination plant (\$/m ³)	1100 [25]
kmo	Water plant owners cost factor (%)	5
kmc	Water plant cost contingency factor (%)	10
cmm	Specific membrane replacement cost (\$/m ³)	0.09
α_{cons}	Drift rate of specific construction cost	0.02
σ_{cons}	Volatility rate of specific construction cost	0.05
α_{cmm}	Drift rate of specific membrane replacement cost	0.01
σ_{cmm}	Volatility rate of specific membrane replacement cost	0.05
r	Discount rate	0.08
AMP	Desalination water plant availability factor	Values for PERT distribution a;b;c 93;95;97
M	Number of Monte-Carlo paths for LMS-MC approach	100,000
Nc	RO desalination plant lead time (yr)	2
No	RO desalination plant lifetime (yr)	20
$W_{selling}^{price}$	Water selling price to the grid, exercise threshold (\$/m ³)	1.9
Wdr	RO Desalination capacity (m3/day)	10,000
dr	The annual decline rate of the unit-generating capacity	0.02

expenditures of desalination plant construction cost ($Cons^{RO}$) which is calculated from the given specific construction cost (SP^{RO}), and overnight capital cost (ON_{cost}). The term “overnight capital cost” is a broad concept encompassing the aggregate of all expenses, both direct and indirect, incurred throughout the pre-construction and construction phases. These expenses include engineering, design, and environmental assessments, land acquisition, site preparation, contract management, insurance, licensing, legal fees, cost adjustments, and contingency provisions throughout the construction period. Then, the annual levelized fixed capital cost (Inv^{RO}) of the RO desalination plant during construction is calculated from the discount rate (r). In contrast, the operational and maintenance expenditures for reverse osmosis (RO), denoted as $O\&M^{RO}$, encompass all annual costs associated with system upkeep. This includes expenses related to power supply, procurement of chemicals and materials for pre- and post-treatment processes, and the replacement of RO membrane modules. To quantify the expenses linked to renewable energy generation, the Levelized Cost of Energy (LCOE) formula is employed. LCOE is conceptually akin to LCOW, representing the annual cost of renewable energy production relative to the quantity of energy harnessed from renewable sources.

$$LCOW = \frac{(Inv^{RO} + O\&M^{RO})}{W_{pms}^{RO}} \quad (14)$$

$$\begin{aligned} CAPEX \\ Cons^{RO} &= SP_{Cons}^{RO} \times Wdr^{RO} \\ ON_{cost} &= Cons^{RO} \times (1 + kmo) \times (1 + kmc) \\ Inv^{RO} &= ON_{cost} \times payper(r, N_c) \end{aligned} \quad (15)$$

$$\begin{aligned} OPEX \\ O\&M^{RO} &= [(cmm \times A) + (LCOE \times 24)] \times W_{pms}^{RO} \\ W_{pms}^{RO} &= W^{RO} \times WPA \times 365 \end{aligned} \quad (16)$$

where N_c is the RO water plant lead time, kdo and kmc are the RO water plant contingency and owner cost factors which cover the uncertainties during the construction period. cmm is the specific cost of replacing membrane cost, A is the membrane area factor, W_{pms} is the annual water production value and is approximated through the water plant availability (WPA).

3.1.2.2. Tax policy. The governmental supporting policies for renewable energy projects comprise both price and tax incentives. In this study, tax expenditure is considered only of value-added and mathematically can be driven as follows:

$$tax = (p^e \cdot q^{NRE} + p^c \cdot q^c) \cdot r^v \quad (17)$$

where p^e and p^c are respectively, the price of renewable energy electricity and CO_2 emissions, q^c is the quantities of CO_2 emission released due to utilizing national grid electricity and r^v denotes the value-added tax.

3.1.2.3. CO_2 price. One significant advantage of substituting conventional fossil fuels with renewable energy lies in the reduction of greenhouse gas emissions. The PV systems, for instance, generate energy directly from sunlight, thereby preventing the release of any greenhouse gases into the atmosphere. Studies indicate that for every 1 kWh generated by fossil fuels, approximately 0.50089 kg of CO_2 is emitted [30]. Consequently, it can be deduced that utilizing conventional fossil fuels to power the RO desalination unit would result in the emission of approximately 6.14 million kg of CO_2 into the atmosphere annually.

In scenario 1, when the RO unit is powered using the national grid, the amount of CO_2 emission (p^c) is considered zero. In Scenario 2, p^c is calculated based on the required energy which is used by the national grid, and finally in Scenario 3 p^c becomes the same as the amount of q^{NRE} .

3.1.2.4. Water selling price. As previously mentioned, the prevailing rate for kilowatt-hour (kWh) costs within subsidies in Saudi Arabia governmental sections stands at 0.085 \$/kWh [31]. It is important to note that the 0.085 \$/kWh price is a subsidized rate limited to governmental facilities in Saudi Arabia. Commercial and industrial users will likely face lower electricity tariff [31]. However, the cessation of governmental subsidies and the deregulation of fossil fuel pricing would necessitate the sale of grid-derived electricity at the international rate of 0.4 \$/kWh. Consequently, the cost of producing one cubic meter of desalinated water would amount to 2.13 \$/m³. This particular variable serves as the critical threshold within the framework of the ROV methodology. This means this price should be the basis for the sale of produced freshwater in all scenarios.

3.2. Power scenario

In this research, we explore three distinct scenarios for RO power supply systems. Firstly, in scenario one, the RO facility is entirely reliant on the national grid for its power needs. In the second scenario, the power generated by the PV plant is allocated exclusively to fuel the RO desalination process, with any surplus power requirements by the RO plant being met by the national grid.

In the third scenario, the RO plant draws its power from both PV energy and an integrated storage system designed to replace grid support. This solar energy storage system consists of batteries capable of storing excess PV energy production, which can be deployed during periods of heightened energy demand when PV generation is insufficient [32,33]. However, environmental impact and high energy footprint, have pushed researchers to work on a new desalination process that is more adapted to nowadays technologies.

Additionally, the RO plant exhibits low energy requirements, thus facilitating load fulfillment during peak hours. In industrial settings, incorporating a battery storage system alongside a PV-grid system may not be economically viable, especially for large-scale applications. As an alternative, the battery storage system serves as a grid substitute, ensuring energy supply during peak load hours when electricity tariffs are at their highest [34]. To compute the annual energy output from PV modules, we propose the following equation:

$$Pv(kWh) = GHI \cdot \eta \cdot f \cdot A_{PV} \quad (18)$$

Here, GHI denotes the annual in-plane global irradiance measured in kWh/m², η represents the efficiency of PV modules, f stands for the performance ratio of PV modules, and A_{PV} signifies the surface area of the PV modules in m². It should be noted that the power generation capacity of the PV power plant (or any other renewable energy generation system) diminishes over time due to natural equipment aging and other associated factors. Consequently, we contemplate three distinct scenarios. The collective outcome is then expressed as follows:

$$GHI_t = GHI_{t-1} \cdot (1 - dr) \quad (19)$$

where dr is the annual declining rate. Based on a hypothetical scenario [26], assuming annual in-plane of 2293 kWh/m², the PV module efficiency of 15 % and the performance ratio of 80 % with PV area of 36,000 m² leads to the PV module annual energy output of 9,811,000 kWh which consists 80 % of the total annual energy requirement for the RO desalination plant to produce 10,000 m³/day of fresh water. The remaining part of the energy requirement will be satisfied by the utility grid.

Furthermore, we assess the influence of implementing a storage bank system on the ultimate water cost. Building upon the preceding scenario, we introduce a battery storage system as a substitute for the grid. This battery storage unit is charged through supplementary PV modules. In the context of this investigation, the battery system, along with the additional PV modules, is designed to offset the energy requirements of the RO desalination plant. Factoring in an efficiency rate of 77 % and a

depth of discharge of 50 % [35], we determine that the surplus energy generated by the PV modules, which must be directed to the battery system, amounts to $(12,226,400 \text{ kWh} - 9,905,760 \text{ kWh} \times 1/0.77 \times 1/0.5) 6,125,300 \text{ kWh}$.

There are two types of battery storage systems; Pb-acid and Lit-ion. In this study, the Pb-acid battery storage system with the leveled cost of storage system of 0.17–0.2 \$/kWh is used.

Drawing upon recent data presented by Ghaithan and colleagues [26], the subsidized electricity tariff prevailing in for governmental sections of Saudi Arabia stands at 0.085 \$/kWh [31] while the LOCE attributed to the PV system is calculated at 0.079 \$/kWh. As a result, the mean electricity cost associated with the PV-grid system is approximated at 0.083 \$/kWh. This proves to be advantageous when compared to the expense of procuring electricity directly from the grid, resulting in annual savings of \$24,528 in energy expenditure. In the context of the last scenario, the average electricity cost amounts to 0.138 \$/kWh.

The acquired LCOW data demonstrates the cost competitiveness of employing renewable-grid-connected systems for desalination when contrasted with reliance solely on the conventional grid. Additionally, it appears that the utilization of a storage system may not offer a compelling advantage in comparison to operating RO desalination systems using the utility grid. The schematic of the proposed LSMC-ROA algorithm is presented in Fig. 4.

4. Results and discussions

This paper considers an exemplary case of an investment project that entails integrating an RO desalination plant with a pre-constructed or pre-operated solar PV power generation plant. The assessment considers the option to defer the investment and the optimum investing time. Further, the benefits of ROA versus traditional appraisal tools, i.e. NPV, are illustrated.

The investment project corresponds to a solar PV-powered RO desalination plant with 10,000 m³/day capacity in Saudi Arabia. To

replicate the conditions found in the most advanced markets, we contemplate three distinct scenarios for energizing the reverse osmosis process. As, fully powered using the national grid; Solar PV and balancing the national grid, self alone solar PV. Each scenario was investigated through two stages:

1. Stochastic modeling provides an evaluation of the economic feasibility by considering market volatilities in the absence of a deferral option and optimizing the time to invest.
2. Optimization of the time to invest where market uncertainties and volatilities are added to model the value of the embedded deferral option.

The investment project's estimated lifespan spans 20 years, commencing from the commissioning date of the desalination plant in question. Furthermore, we contemplate the possibility of a deferral period lasting up to 7 years, beyond which the building permit for the desalination plant expires. Should the investor opt for project deferral, it results in a delay of the initial capital outlay and the forfeiture of the revenue stream that the project would otherwise generate immediately post-commissioning. The risk-free discount rate, or the capital cost utilized for computing the option-to-defer value and the Monte Carlo-based semi-new appraisal of NPV, is set at 8 %.

The Results and discussions section is arranged in two main parts: the presentation of the results from the modeling of the RO system and the subsequent presentation of the results from the modeling of the real options approach.

In continue, the preliminary results to elucidate the final cost outcomes of freshwater production are presented. The modeling outcomes reveal that the LCOW parameter values for scenarios 1, 2, and 3 are 1.47 \$/m³, 1.43 \$/m³, and 1.91 \$/m³, respectively, Scenario 2 exhibits the lowest cost for freshwater production, attributed to the utilization of solar energy and its effect on achieving a lower LCOW. Conversely, in scenario 3, which incorporates battery usage, the leveled cost per

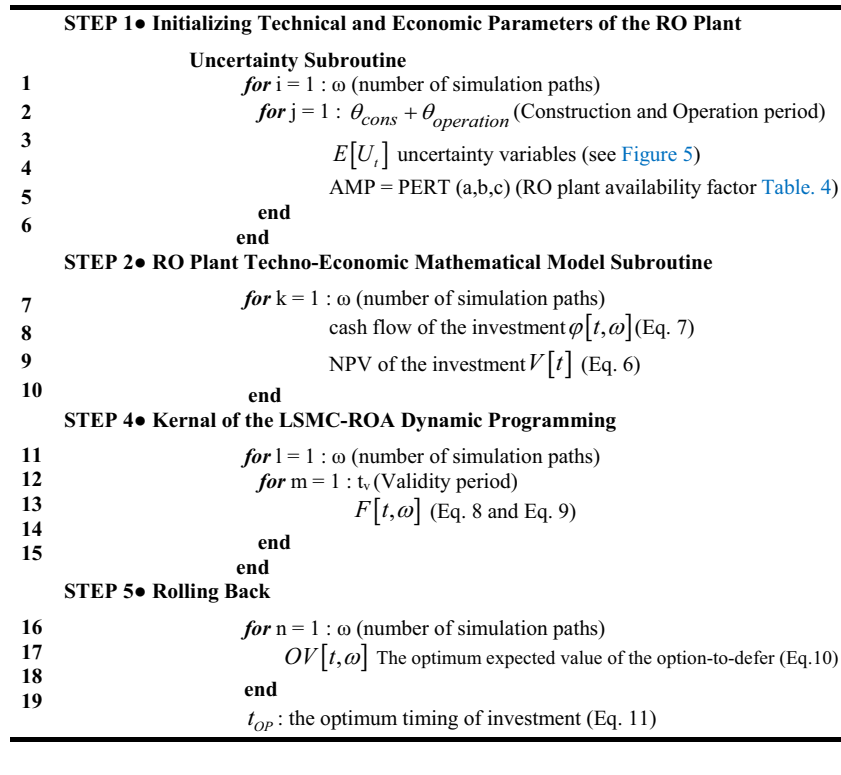


Fig. 4. LS-MC procedure in estimating options value and optimum investing time.

Table 5

Analysis of LCOW key contributing factors in different Scenarios.

		Construction cost (M\$)	Owners cost (M\$)	Contingency cost (M\$)	Annual O&M cost (M\$)	Annual electricity cost (M\$)	LCOW (\$/m ³ /day)
Scenario 1	Fully grid	11.0	0.55	1.1	0.47	2.6	1.47
Scenario 2	PV-grid					2.7	1.43
Scenario 3	PV-Battery					4.1	1.91

kilowatt-hour of solar energy surpasses the other two scenarios. Additionally, noteworthy is the average freshwater production across various scenarios. Table 5 examines the components comprising the LCOW.

The next part involves comparing the overall investment value using classical methods and flexibility in making investment decisions to demonstrate the strength of the Real-Option Theory in identifying investment opportunities under high uncertainty. This is particularly important in situations where classical appraisal methods deem investments as economically unviable. We examine the investment value and optimal investment timing for solar PV-powered desalination in Saudi Arabia.

Under the classical investment evaluation techniques, the NPV of the investment project without flexibility is calculated as the difference between the expected present value of the revenues (Rev) and investment costs associated with the project (Exp). The revenues generated by the desalination project are a function of the amount of selling produced fresh water.

In the Monte Carlo analysis of the Net Present Value, we estimate the anticipated present value of revenues ($E[Rev]$) and the expected value for expenditures ($E[Exp]$) through Monte Carlo simulations. This entails generating a substantial number of potential scenarios by stochastically sampling market variables from their respective probability distributions over the lifespan of the investment project. For each simulated scenario, we compute the present value of cash flows. Subsequently, by iteratively performing this procedure numerous times, we derive a distribution of NPV values, enabling the determination of metrics like the anticipated NPV, as outlined below:

$$NPV_{Monte-Carlo} = E[Rev] - E[Exp] \quad (20)$$

For the exemplary case analyzed, the expected NPV for different power options in scenarios 1 to 3 of the project is calculated as -13.2 M\$ for Scenario 1, 11.8 M\$ for Scenario 2, and -31.2 M\$ for Scenario 3.

Using the classic NPV assessment, the first and third scenarios of the investment project may result in rejecting it. Because the investment project does not create value for the investors ($NPV < 0$). Using only traditional appraisal methods could lead investors to miss out on investment opportunities. Relying solely on solar energy for RO desalination may not be cost-efficient for producing freshwater. This point is of great significance.

Nevertheless, when contemplating the potential to delay investment, irreversible projects, such as those involving renewable energy desalination, should not be summarily dismissed. These projects may indeed present appealing prospects for investors, especially when uncertainties can be, at least partially, resolved. As previously discussed, the value of the embedded options within an investment project hinges on the new information that may materialize in subsequent years. The deferral option enables a 'wait-and-see' approach, wherein a decision to proceed with the project is deferred until uncertainties are favorably resolved. Conversely, if future conditions turn unfavorable, the investor allows the option to expire, resulting in the non-execution of the renewable desalination project. In this manner, a positive investment value for the project can emerge, revealing an attractive investment opportunity. Consequently, the revenue generated by the embedded deferral option, as determined through LSMC analysis, amounted to $\$57.2$ million for the first scenario, $\$47.7$ million for the second scenario, and $\$61.7$ million for the third scenario. The incorporation of flexibility, or real options,

can significantly increase the economic value of investment projects. The value of a project with flexibility is computed as the Monte-Carlo derived NPV plus the economic value of the embedded options.

$$FLV = NPV_{Monte-Carlo} + RV \quad (21)$$

The total investment value with the flexibility to defer or wait for new information through Eq. (14) would be 43.9 M\$ (-13.2 M\$ + 57.2 M\$), 59.6 M\$ (11.8 M\$ + 47.7 M\$), 30.5 M\$ (-31.2 M\$ + 61.7 M\$). However, the optimal decision is not to invest immediately. The optimal investment time is the second year of the validity period.

In scenario 1, since the required energy is supplied by the national grid, CO₂ taxation should be considered as part of the expenses and be paid. However, CO₂ tax is part of the income in scenarios 2 and 3, which rely on solar energy.

When Carbon emission trading is not considered, the total investment values for scenarios 2 and 3 would be 56.04 (6.48 + 49.56) M\$ and 30 (-32.7 + 62.8) M\$, respectively. It can be observed that in the absence of CO₂ taxes, the overall investment value in scenario 2 experiences a negligible reduction due to a decrease in the value of real option value. However, in this case, the NPV remains positive. In scenario 3, the overall investment value also undergoes. Based on the classical evaluation of NPV, this project is still deemed unfavorable.

Here, the robustness of the developed LSMC technique should be examined. The robustness of the LSMC assessment model is determined through the computation of the deferral option's value across an extensive array of simulated pathways for the variables. Fig. 5 illustrates respectively 10,000 samples of simulated specific construction cost of RO plant. The specific operation and maintenance expenditure during the operation period was assumed to follow a similar behaviour pattern and for AFWP, we utilized the PERT statistical distribution to account for its inherent variability.

Fig. 6 illustrates the statistical convergence of the differing option values on the number of Monte-Carlo realizations/simulation paths. It

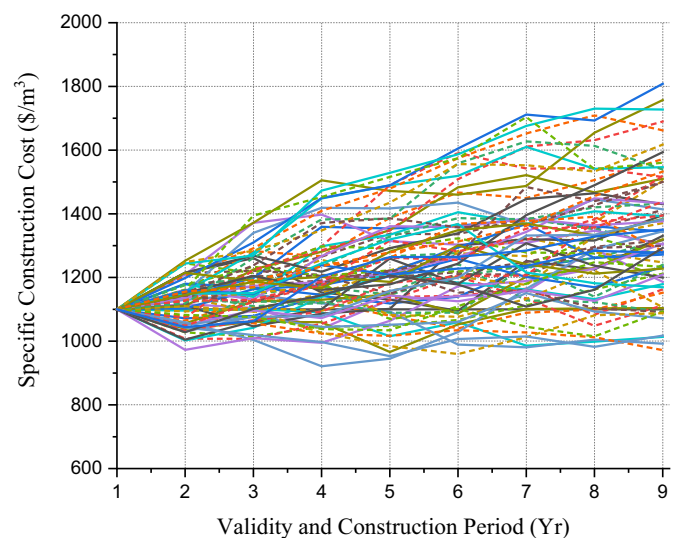


Fig. 5. Realizations of specific investment cost during validity (tv) and construction period (Nc).

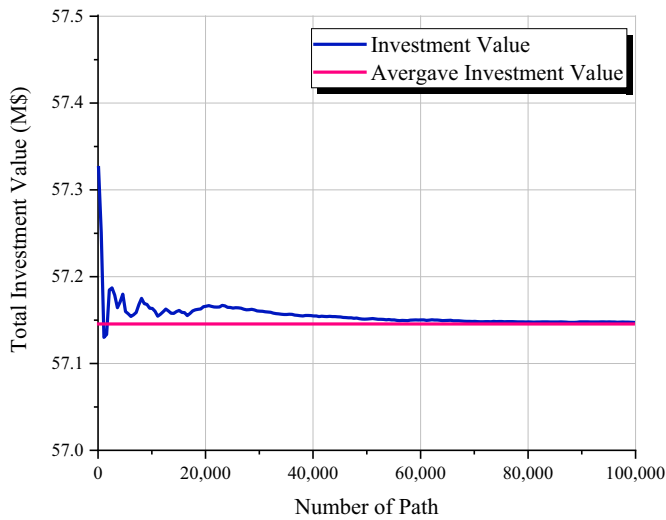


Fig. 6. Total project investment value with flexibility according to the number of Monte-Carlo generation.

can be seen that the solution is robust for the number of simulation paths $>40,000$ number.

4.1. Analysis of the parameters influencing the value of the deferral options

Once the value is provided by flexibility for the embedded options in the renewable desalination investment, it becomes feasible to visually depict various decision domains. These domains are contingent upon specific parameters that exert an influence on both the options' worth and the NPV. Table 6 delineates the criteria governing each of these decision domains. To define these decision domains accurately, it is imperative to calculate the values of flexible options (RV), Monte Carlo-derived NPV, and the total investment outlay across varying input variables.

From the conditions specified in Table 6, a sensitivity analysis for the parameters influencing the value of the deferral options can be carried out. For this purpose, the effect of various economic and technical parameters is investigated. These parameters are selected mainly the salinity of feedwater, desalination plant capacity, specific desalination plant construction, selling water price, weighted average capital cost (WACC), and maturity period.

Fig. 7 illustrates the decision-making regions according to the variations in the concentration of the feed water (X_f). In all scenarios a significant increase in the NPV object, for X_f higher than 45,000 ppm is observed, while simultaneously leading to a decrease in the option value. Consequently, the total investment value which represents the sum of these two factors, remains relatively constant. Hence, it can be inferred that the feedwater salinity, as a parameter influenced by site-specific characteristics, exerts a substantial impact on the final

Table 6
Decision rules of decision-making.

Option type	Conclusion	State I	State II	State III
Option to Defer	Defer investment	$NPV < 0$	$RV > 0$	$FLV > 0$
	Invest now	$NPV \geq 0$	$RV = 0$	–
	Refuse investment	$NPV < 0$	$RV > 0$	$FLV < 0$
Option growth or abandon	With options	$NPV \geq 0$	$RV > 0$	$FLV > NPV$
	Without options	$NPV \geq 0$	$RV = 0$	–
	Refuse Investment	$NPV < 0$	$RV > 0$	$FLV < 0$

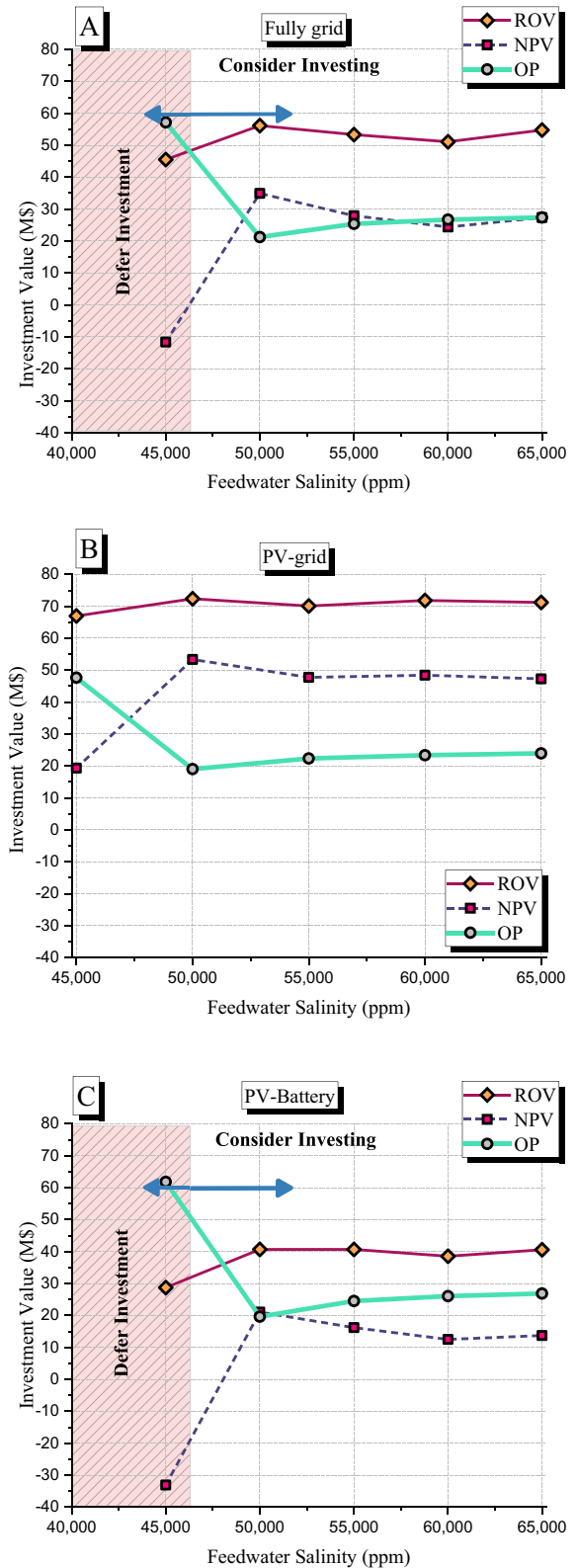


Fig. 7. Total project investment value according to feedwater salinity of RO desalination system at the second year of maturity period A) fully grid B) PV-grid C) PV-Battery.

investment decision process.

In scenarios 1 (Fully grid) and 3 (PV-battery), two discernible trends emerge: the choice to postpone investment and the choice to proceed with investment, irrespective of the project's unfavorable NPV. Despite the consistently positive option value, it remains insufficient for certain parameters, resulting in a negative flexible NPV. Conversely, in scenario 2 (PV-grid), where the flexible NPV is positive, the investment decision can be made without constraints.

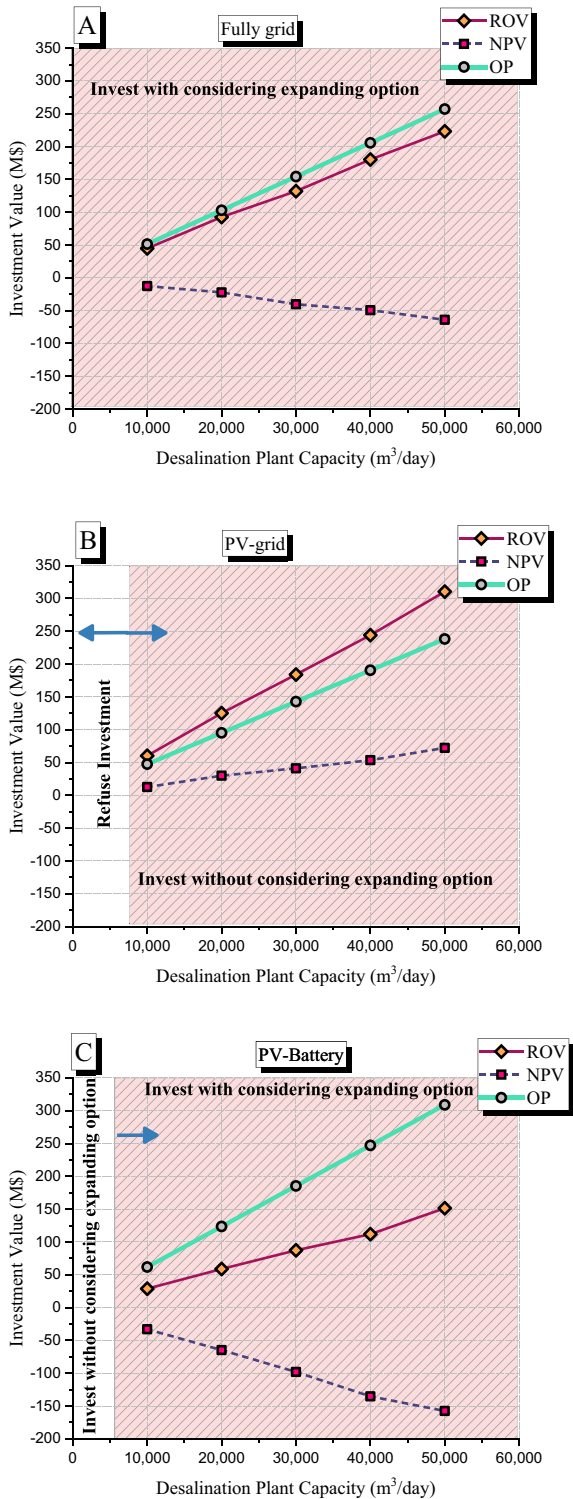


Fig. 8. Total project investment value according to RO desalination system capacity A) fully grid B) PV-grid C) PV-Battery.

Fig. 8 investigates the effect of freshwater production capacity on the overall investment value. Increasing the freshwater production capacity requires the expansion of the existing desalination unit. Hence, the option for development and expansion could be considered. This option refers to the potential for capital investment during the construction or operation phase of the project to achieve higher profits.

In this manner, in scenario 1 (Fully grid) by keeping open the option to growth while the risk of losses is limited, a higher probability of increasing profits could be achieved. This is because the volatility rate of parameters that are related to investing in renewable desalination plants could lead to higher income. However, in scenarios 2 (PV-grid) and 3 (PV-Battery), the two-decision region of refusing the investment and the decision to increase the capacity regardless of the value of the option value are evident.

The decision-making according to the weighted average capital cost (WACC) and the initial investment cost of the desalination plant is shown in Figs. 9 and 10, respectively.

The intrinsic value of an investment project's underlying asset corresponds to the current valuation of cash flows generated by the project. This valuation is intrinsically linked to the discount rate or the risk-adjusted capital cost, often denoted as WACC. When the capital cost escalates, it is anticipated that the intrinsic value of the underlying asset will diminish, and conversely, a reduction in the capital cost is expected to elevate this intrinsic value. Notably, the option to defer, akin to a financial call option, experiences a reduction in its worth as the intrinsic asset value decreases, particularly in tandem with an escalation in the risk-adjusted capital cost.

This highlights the importance of analyzing the components that contribute to the total investment value concurrently. From the provided decision-making rules; for the second scenario, two regions of the decision to defer the investment and the decision to invest are observed regardless of the value of the flexible NPV and the value of the deferral option. However, scenarios 1 (Fully grid) and 3 (PV-battery) reveal the potential to keep open the deferral option Even though the flexible NPV is negative in some situations.

The impact of the risk-adjusted capital cost WACC on the deferral option value for financing renewable desalination projects is illustrated in Fig. 10. In scenarios 1 (Fully-grid) and 2 (PV-grid), it is evident that by maintaining the option to defer, the escalation in costs surpasses the increase in returns from the desalination project. Consequently, the option value diminishes, prompting an immediate investment decision. In scenario 3 (PV-Battery), where the energy required for desalination is provided by a self-contained PV plant operating independently of the grid and storage systems, the project proves to be financially viable. In this case, the decision to invest may be postponed or declined, contingent upon the cost of capital and the initial investment outlay under consideration.

To comprehend the evolving worth of the option as maturity time extends, a sensitivity analysis was conducted while maintaining a constant risk-adjusted capital. The findings derived from this analysis illustrate that as the expiration date prolongs, the option's value escalates, as illustrated in Fig. 11. This phenomenon can be elucidated by the fact that investors are afforded more time to await valuable insights and the resolution of pivotal uncertainties, thereby enabling more informed decision-making. If the option to postpone the construction of the desalination plant has no expiration, it sends an enduring economic signal to investors, encouraging perpetual deferment of construction due to the persistent growth in the option's value year by year.

The option to defer, equivalent to a financial call option, decreases its value with a decrease of the underlying asset value, in our case, with the increasing water selling price. Fig. 12 illustrates the value of the underlying asset under the option to defer concerning the annual escalation of the water selling price. If the value of the option to defer is low, the investor will avoid postponing the investment in the project and the investment will be executed as soon as possible. It is noted that in all scenarios the deferral option value remains constant. But keeping open

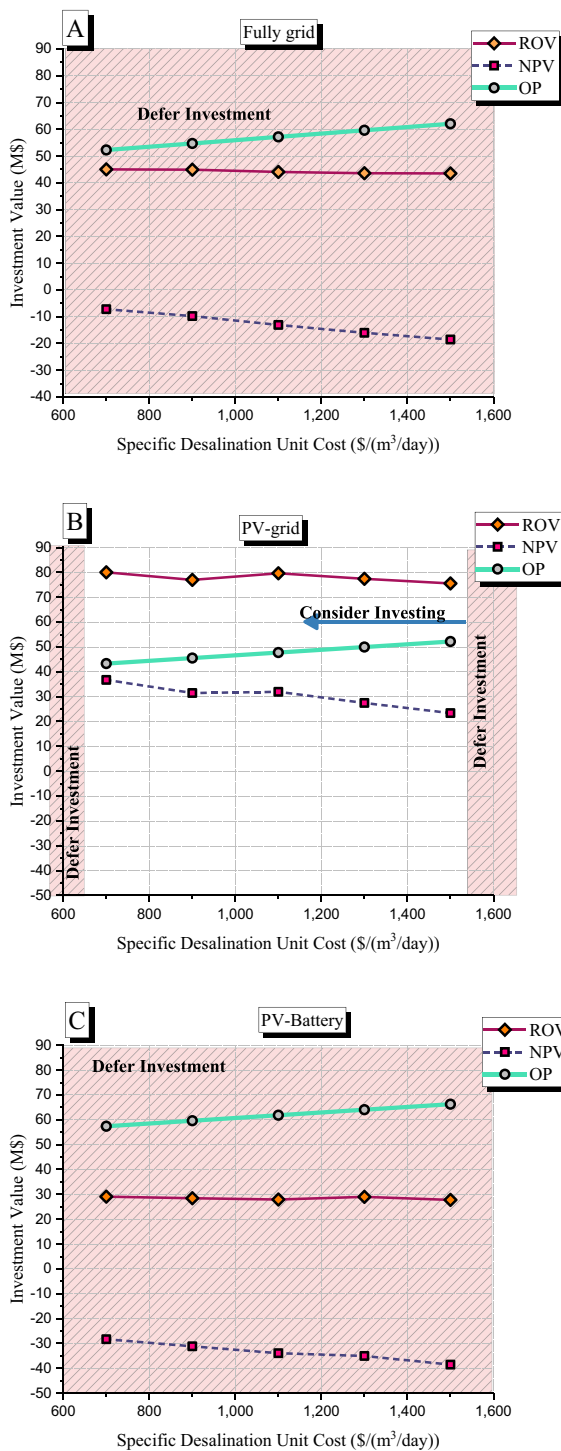


Fig. 9. Total project investment value according to specific desalination unit construction cost A) fully grid B) PV-grid C) PV-Battery.

the option to defer a higher probability of increasing profits could be achieved.

5. Conclusion

In light of the escalating demand for freshwater in arid and semi-arid regions, particularly in the Middle East, where resources are becoming increasingly scarce, the significance of sustainable solutions is paramount. Seawater desalination presents a promising prospect in areas with access to seas. However, seawater desalination is a highly energy-

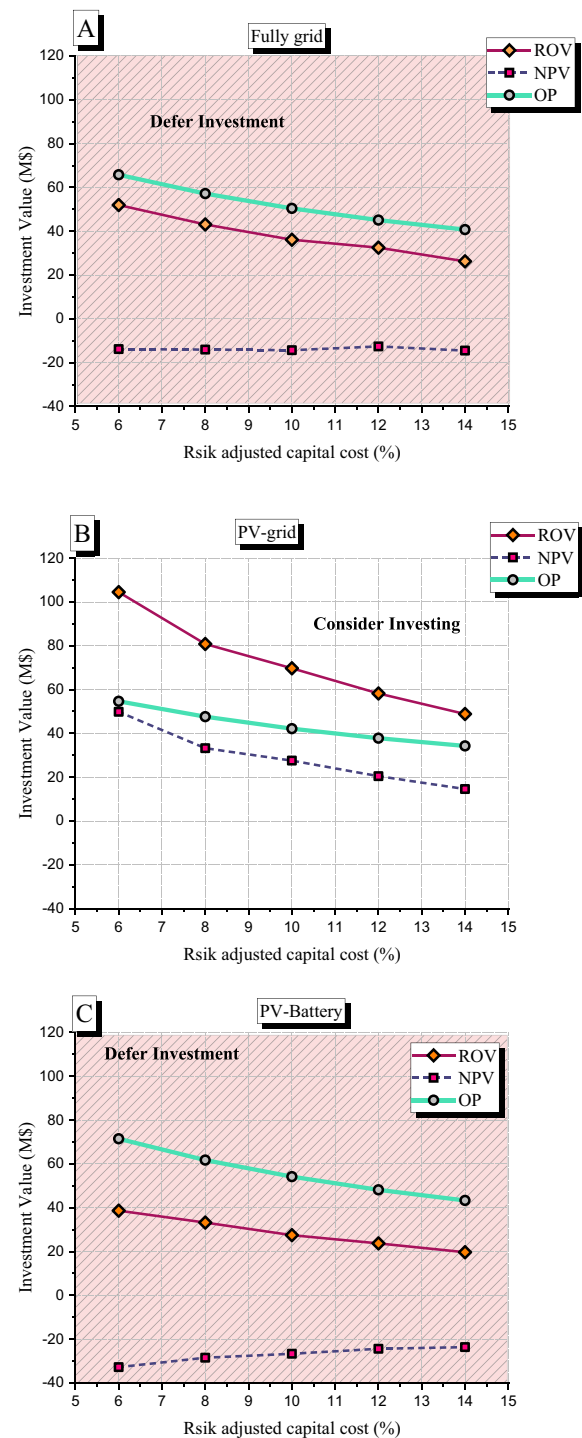


Fig. 10. Total project investment value according to discount rate variation A) fully grid B) PV-grid C) PV-Battery.

intensive process. Furthermore, aside from the rapid increase in efficiency of desalination technologies, providing a clean energy supply as well as supporting sustainable economic development should be carefully examined. One of the crucial aspects of the feasibility study of utilizing renewable energy for desalination is to consider risks arising from uncertainties such as the cost of the required energy. In this paper, a novel decision-making numerical tool to evaluate the opportunities in renewable PV-RO water desalination, by introducing the Least Square Monte-Carlo Real Options framework to achieve the objectives is presented for pursuing two objectives:

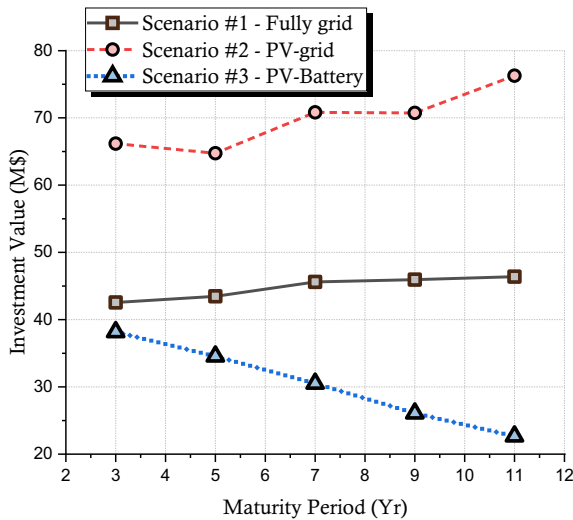


Fig. 11. Total project investment value according to maturity period variation.

- (1) addressing the uncertainties in the renewable-energy-powered membrane desalination projects, specifically RO desalination plants powered by photovoltaic power plants (PV-RO).
- (2) Evaluating the value of flexibility of decision-making regarding investments in seawater desalination using renewable energy (PV-RO). Therefore, decision-makers could decide whether or not to invest in every stage, as long as the uncertainties could be manageable.

While traditional appraisal techniques using Discounted Cash Flow (DCF) evaluation might immediately reject certain cases, the Real Options Analysis (ROA) methodology suggests that the option-to-defer value can expand as flexibility investment value increases. This inherent advantage of ROA techniques facilitates a comprehensive evaluation of deferring project funding, rather than prematurely discontinuing investments. Therefore, it is expected that this framework could help investors and decision-makers to assess seawater desalination powered by renewable energy projects with high volatility and risk.

Furthermore, a graphical presentation depicting the different areas of decision-making rules based on the determined values of the embedded options is suggested. These areas are a function of certain parameters that influence the option's value and NPV. To illustrate the theoretical and practical differences between traditional appraisal methodologies and advanced ROA techniques an exemplary case study regarding an RO desalination plant coupled with a pre-constructed photovoltaic power plant in KSA was used.

The authors considered three pricing scenarios for powering the RO plant:

- In scenario one (fully grid), the RO plant will be fully powered using the national grid.
- In scenario two (PV-grid), the PV plant is supposed to be powered by the RO desalination process, and the national grid is used to balance any excess power requirement.
- In scenario three (PV-Battery), the RO plant is dependent on a PV plant and a battery storage system to substitute the national grid.

To maximize the profit of desalination renewable energy projects in Middle Eastern countries, decision-makers could take into account the following order of priority: seawater salinity, desalination plant capacity, LCOE price, desalination plant's specific construction (Inv) and O&M cost, risk-adjusted capital cost (WACC) and maturity period of investment (t_v). The results showed that in the absence of managerial flexibility and flexible investment timing, the typical DCF technique suggests

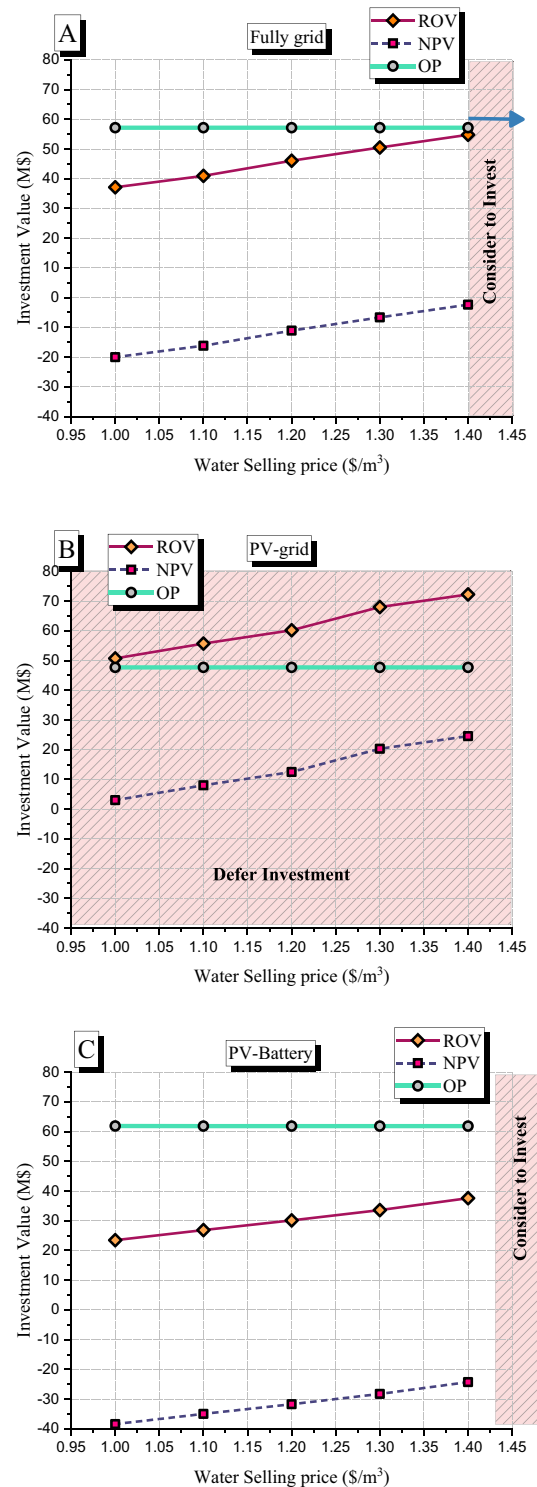


Fig. 12. Total project investment value according to water selling price A) fully grid B) PV-grid C) PV-Battery.

the lowest total investment value, while, with the proposed advanced LSMC-ROA methodology, a higher profit could be expected. The sensitivity analysis is a proper tool that potentially could highlight the influence of uncertain factors on the value of the deferring exercise of the options. In continuation, the effect of the mentioned uncertainty factors on the total deferral option value is presented:

- A threshold salinity level for feedwater exists (45,000 ppm) and when the salinity of the feedwater is lower than this, the traditional appraisal DCF technique proves it would not be economically favorable.
- Further, it was shown that by increasing the specific cost of the desalination unit, the level of the deferral option value increases too. However, the expected NPV value of the investment decreases due to a significant increase in initial investment.
- The impact of escalation of the risk-adjusted capital cost on the expected NPV value shows a different trend for scenarios two (PV-grid) and three (PV-Battery). In scenario three (PV-Battery) increasing the risk-adjusted capital cost ratio could turn the investment into an economically attractive project, while when the national grid is supposed to balance the excess power requirement of the desalination unit, the expected value of NPV would become negative, and turns the investment uneconomic.
- Next, the effect of extending the validity period on the total flexible investment value under different power pricing scenarios has shown that under scenario two (PV-grid) the total investment increases while under the third scenario (PV-Battery) this value decreases.
- Finally, the influence of the selling water price's uncertainty showed that the greater the water price volatility and drift rate, the higher the option value could be expected.

In conclusion, the proposed LSMC-ROA framework could effectively address uncertainties and assess economic feasibility using the ROA technique. Thus, both objectives of the study have been successfully achieved. However, more precise quantification of project uncertainties remains a future research priority for enhanced risk management in renewable energy desalination projects across the Middle East. As uncertainties associated with renewable energy and regional conditions evolve due to factors like climate change, environmental policies, and economic conditions, the ROA methodology should continue evolving. Strengthened by subsidies, precise power scenarios, or big data analysis, the proposed model is poised to substantially enhance the value of renewable seawater desalination investments in Middle Eastern countries.

CRediT authorship contribution statement

Pouyan Najafi: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammadreza Aghaei:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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